The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 3 MODULE, AUTUMN SEMESTER 2010-2011

COMPUTER VISION

Time allowed TWO Hours

Candidates must NOT start writing their answers until told to do so

Answer THREE out of FIVE questions

No calculators are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

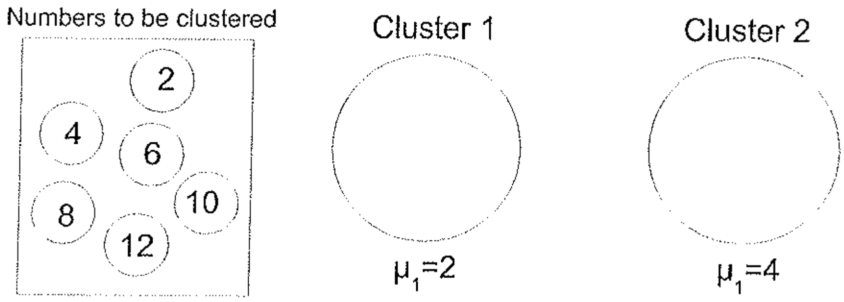
No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

1. Answer all of the following sub-questions pertaining to Segmentation (25 marks):

   (a) Use the k-Means clustering method to cluster the numbers depicted in the diagram below (i.e. 2, 4, 6, 8, 10 and 12). Assume that the number of clusters is two, and that the means (i.e. μ) are initialized to 2 and 4. For your answer, depict 4 iterations of the method. Note that the calculations involved are simple enough to preclude the need for a calculator.

   [7 marks]



   (b) Briefly discuss four different issues (or problems) that may be faced when applying the k-Means clustering method.

   [4 marks]

   (c) Discuss potential solutions to the issues you outlined in your previous answer and discuss known extensions to the basic method. In total you should discuss four solutions and/or extensions (e.g. "2 solutions and 2 extensions", or "3 solutions and 1 extension", etc.).

   [6 marks]

   (d) Describe at length two different region-based segmentation methods. Focus mainly on the algorithmic steps of both methods.

   [8 marks]

2. Answer all of the following sub-questions pertaining to Binocular Stereo (25 marks):

(a) Explain the concept of disparity and how it contributes towards the computation of depth.

[4 marks]

(b) Different Computer Vision applications call for different camera setups. Briefly describe three characteristics according to which camera setups may differ. Explain why camera calibration is necessary.

[5 marks]

(c) Do humans require high-level monocular cues for solving the binocular stereo problem? Justify your answer. How does your answer influence the way you might design a stereo algorithm?

[5 marks]

(d) What is the correspondence problem? How can camera geometry make the problem easier to solve?

[7 marks]

(e) What conditions favor the formation of dense or sparse disparity maps? What are the pros and cons of each one (i.e. dense vs. sparse disparity maps)?

[4 marks]

3. Answer all of the following sub-questions pertaining to Motion Analysis and Optic Flow (25 marks):

   (a) Why is Motion Analysis an important and useful visual function? Provide at least four different reasons.

   [4 marks]

   (b) Briefly discuss three different reasons why Motion Analysis is a difficult problem.

   [3 marks]

   (c) Assume that you were hired to build a Computer Vision system for detecting a burglar breaking into a garage. The setup consists of a single camera inside the garage, with a complete view of all possible entry points. Assume that the lighting inside the garage can be affected by the external lighting (e.g. sun, moon, car lights, etc.). Your solution should be based on a simple motion detection approach.

      i. Briefly explain two problems your motion detection solution is likely to face.

      [4 marks]

      ii. Provide a sketch of the algorithm you would use for motion detection and explain how your algorithm takes care of the problems you outlined.

      [5 marks]

   (d) Explain and differentiate the concepts of *Motion Field* and *Optical Flow*.

   [4 marks]

   (e) What is the *Aperture Problem* and how can it be addressed?

   [5 marks]

4. Answer all of the following sub-questions pertaining to Tracking (25 marks):

(a) Visual trackers consist of three main components, i.e.: motion model, appearance model and tracking engine/framework. Briefly explain the meaning of each one of these components.

[3 marks]

(b) One way to track objects is by attempting to match whole image-patches. Discuss some of the issues faced by this approach.

[4 marks]

(c) Discuss at length four of the main general issues faced by trackers.

[6 marks]

(d) Active contours are typically controlled by an energy function that consists of both internal and external energy components. Give mathematical expressions for these components, and for the complete energy function, and explain how the function guides an active contour during tracking.

[8 marks]

(e) Consider again tracking based on active contours. Assume that you are tracking simple shapes that differ from each other in terms of basic properties such as the number and relative positioning of corners (e.g. triangle vs. pentagon). Assume also that you don't know beforehand what shape needs to be tracked at any particular time. Propose a simple modification of the basic active contours approach that can use the basic shape information mentioned above, in order to track more effectively.

[4 marks]

5. Answer all of the following sub-questions pertaining to Event Detection (25 marks):

   (a) Explain the two terms *Hidden* and *Markov* in the expression Hidden Markov Models.

   [4 marks]

   (b) A Hidden Markov Model consists of two sets and three functions. What are they?

   [5 marks]

   (c) List and briefly explain three questions that Hidden Markov Models allow us to ask regarding a set of observations.

   [3 marks]

   (d) Imagine a car park scenario, where individuals can be in one of four states: 1) walking, 2) standing, 3) entering/exiting a car or 4) driving. Draw a state transition diagram corresponding to this scenario, with plausible estimates of the starting and transition probabilities between the states.

   [6 marks]

   (e) Consider again the scenario described in sub-question (d). Provide an explanation of how you might use Hidden Markov Models in order to detect suspicious behaviours. A suspicious individual might for example walk to the car park, then stand, then walk, then enter/exit a car, then stand, then walk, then stand again, and so on. Note that there are many other examples of suspicious behaviour. In your explanation, make sure you include plausible examples of the observations that may stem from different states.

   [7 marks]